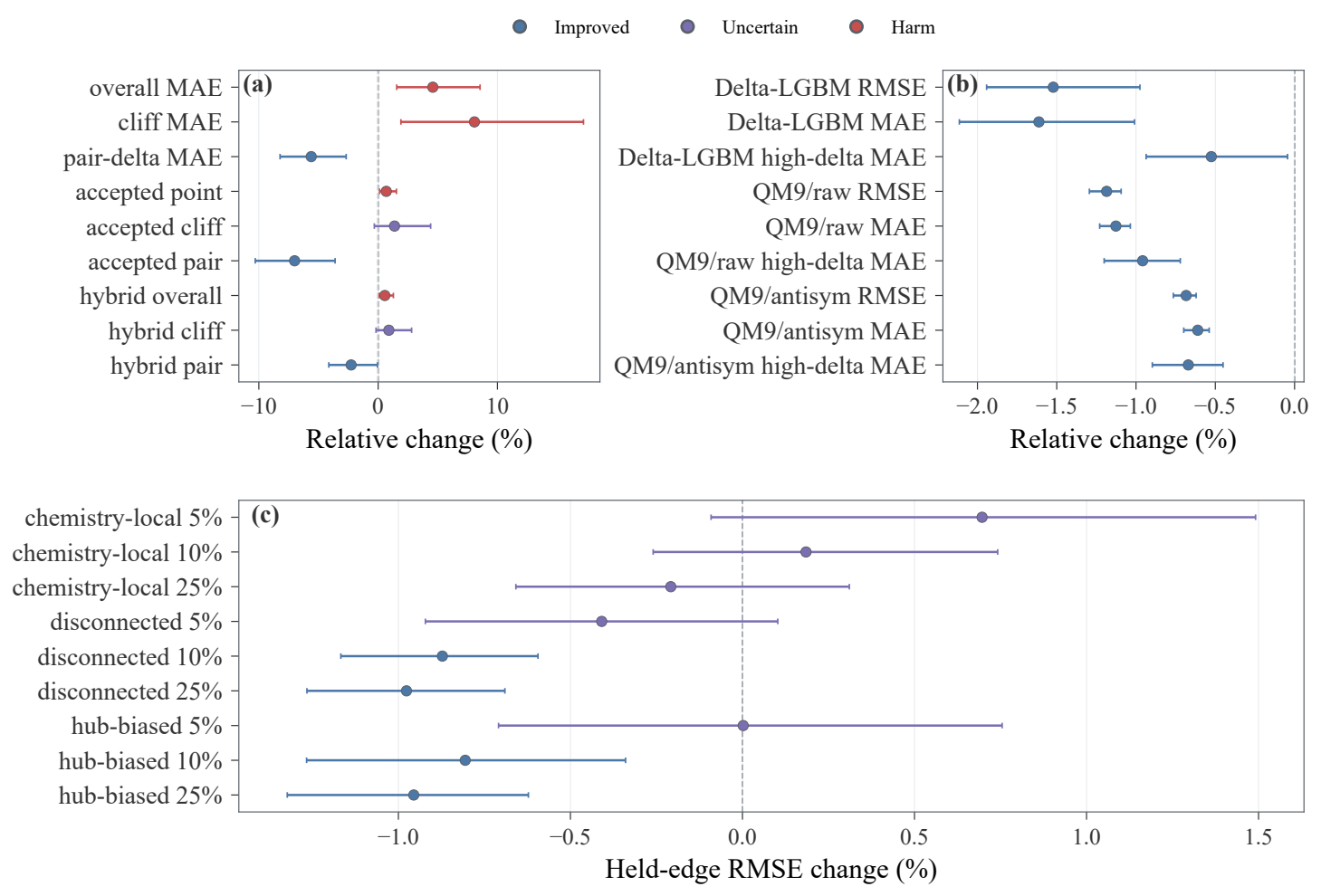
Improved
Uncertain
Harm
(a)
overall MAE
cliff MAE
pair-delta MAE
accepted point
accepted cliff
accepted pair
hybrid overall
hybrid cliff
hybrid pair
−10
0
10
Relative change (%)
(b)
Delta-LGBM RMSE
Delta-LGBM MAE
Delta-LGBM high-delta MAE
QM9/raw RMSE
QM9/raw MAE
QM9/raw high-delta MAE
QM9/antisym RMSE
QM9/antisym MAE
QM9/antisym high-delta MAE
−2.0
−1.5
−1.0
−0.5
0.0
Relative change (%)
(c)
chemistry-local 5%
chemistry-local 10%
chemistry-local 25%
disconnected 5%
disconnected 10%
disconnected 25%
hub-biased 5%
hub-biased 10%
hub-biased 25%
−1.0
−0.5
0.0
0.5
1.0
1.5
Held-edge RMSE change (%)